

# torch.quantize\_per\_channel#

[https://pytorch.org/docs/stable/generated/torch.quantize\\_per\\_channel.html](https://pytorch.org/docs/stable/generated/torch.quantize_per_channel.html)

## Description:

Converts a float tensor to a per-channel quantized tensor with given scales and zero points. input (Tensor) – float tensor to quantize scales (Tensor) – float 1D tensor of scales to use, size should match input.size(axis) zero\_points (int) – integer 1D tensor of offset to use, size should match input.size(axis) axis (int) – dimension on which apply per-channel quantization

## Function Signature

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```
torch.quantize_per_channel(input, scales, zero_points, axis, dtype) → Tensor#
```

Parameters

Returns

Return type

### Returns:

Tensor

## Code Examples

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```
>>> x = torch.tensor([[ -1.0, 0.0], [ 1.0, 2.0]])
>>> torch.quantize_per_channel(x, torch.tensor([0.1, 0.01]),
torch.tensor([10, 0]), 0, torch.quint8)
tensor([[ -1.,  0.],
        [ 1.,  2.]], size=(2, 2), dtype=torch.quint8,
        quantization_scheme=torch.per_channel_affine,
        scale=tensor([0.1000, 0.0100], dtype=torch.float64),
        zero_point=tensor([10,  0]), axis=0)
>>> torch.quantize_per_channel(x, torch.tensor([0.1, 0.01]),
torch.tensor([10, 0]), 0, torch.quint8).int_repr()
tensor([[ 0, 10],
        [100, 200]], dtype=torch.uint8)
```

# Detailed Documentation

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## Documentation

Converts a float tensor to a per-channel quantized tensor with given scales and zero points.

input (Tensor) – float tensor to quantize

scales (Tensor) – float 1D tensor of scales to use, size should match input.size(axis)

zero\_points (int) – integer 1D tensor of offset to use, size should match input.size(axis)

axis (int) – dimension on which apply per-channel quantization

dtype (torch.dtype) – the desired data type of returned tensor. Has to be one of the quantized dtypes: torch.quint8, torch.qint8, torch.qint32

A newly quantized tensor